

# Artificial Intelligence

## First-Order Logic

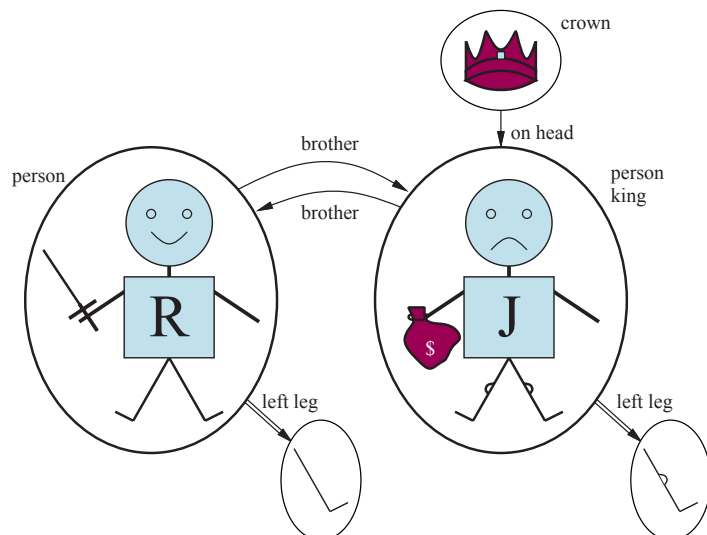
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### 1 Representation

| Language            | Ontological Commitment<br>(What exists in the world) | Epistemology<br>(What are we trying to know) |
|---------------------|------------------------------------------------------|----------------------------------------------|
| Propositional logic | facts                                                | true/false                                   |
| First-order logic   | facts, objects, relations                            | true/false                                   |
| Temporal logic      | facts, objects, relations, times                     | true/false                                   |
| Probability theory  | facts                                                | degree of belief                             |
| Fuzzy logic         | facts with degree of truth $\in [0, 1]$              | known in advance                             |

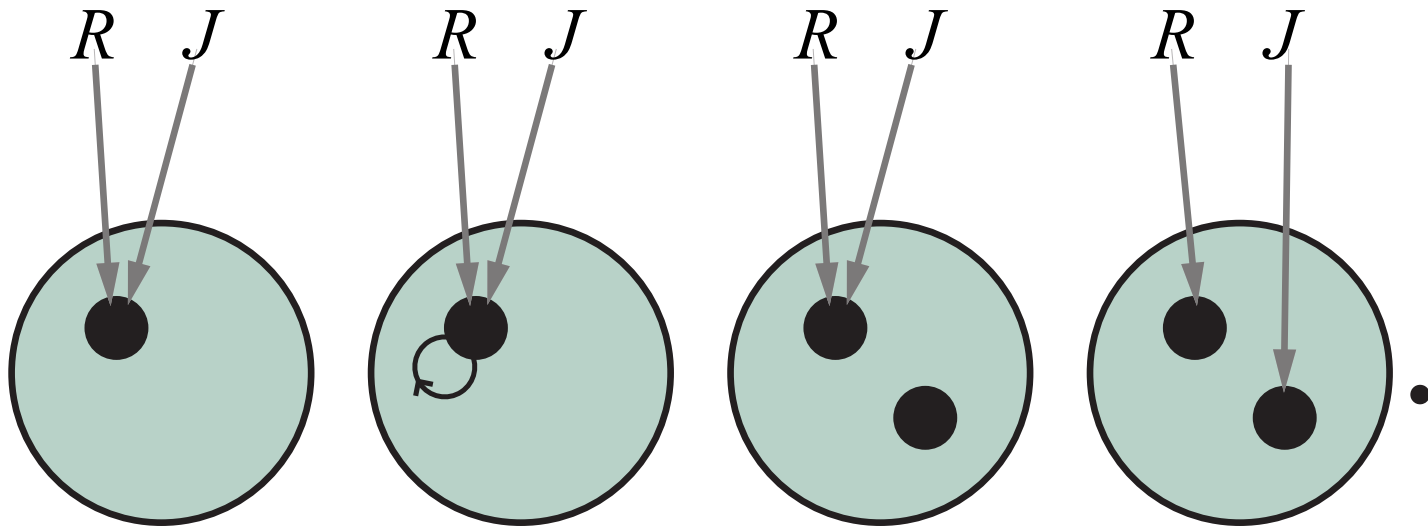
### 2 Syntax and Semantics of First-Order Logic



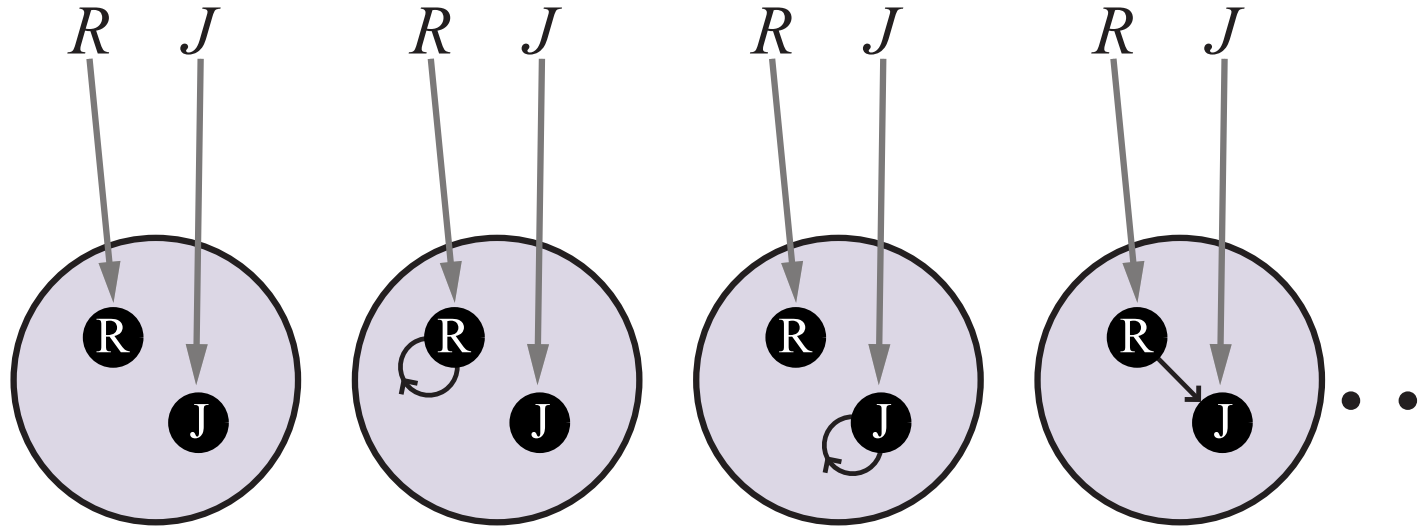
### 3 Syntax and Semantics of First-Order Logic

$Sentence \rightarrow AtomicSentence \mid ComplexSentence$   
 $AtomicSentence \rightarrow Predicate \mid Predicate(Term, \dots) \mid Term = Term$   
 $ComplexSentence \rightarrow (Sentence)$   
 $\quad \mid \neg Sentence$   
 $\quad \mid Sentence \wedge Sentence$   
 $\quad \mid Sentence \vee Sentence$   
 $\quad \mid Sentence \Rightarrow Sentence$   
 $\quad \mid Sentence \Leftrightarrow Sentence$   
 $\quad \mid Quantifier Variable, \dots Sentence$   
  
 $Term \rightarrow Function(Term, \dots)$   
 $\quad \mid Constant$   
 $\quad \mid Variable$   
  
 $Quantifier \rightarrow \forall \mid \exists$   
 $Constant \rightarrow A \mid X_1 \mid John \mid \dots$   
 $Variable \rightarrow a \mid x \mid s \mid \dots$   
 $Predicate \rightarrow True \mid False \mid After \mid Loves \mid Raining \mid \dots$   
 $Function \rightarrow Mother \mid LeftLeg \mid \dots$   
  
 OPERATOR PRECEDENCE :  $\neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow$

### 4 Syntax and Semantics of First-Order Logic



## 5 Syntax and Semantics of First-Order Logic



## 6 Using First-Order Logic

Foo

## 7 Knowledge Engineering in First-Order Logic

